Arista Access Node Hardware/Environmental Fault Recovery Procedure for Hebbal Edge

Node: HEB-ACC-63 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9074

Description

The occurrence of hardware or environmental faults, such as ASIC memory parity errors, line card crashes, or power supply redundancy loss, within the Arista access node HEB-ACC-63 at the Hebbal Edge site, triggers a major alarm (ALARM_9074) necessitating immediate attention. This failure domain is often precipitated by underlying issues within the node's hardware components or its environmental operating conditions. Specifically, the ASIC (Application-Specific Integrated Circuit) memory parity errors can arise from defects in the manufacturing process, wear and tear over time, or exposure to environmental stressors such as high temperatures or electrical surges. Line card crashes may be attributed to software glitches, hardware malfunctions, or the aforementioned environmental factors. The loss of power supply redundancy, critical for ensuring continuous operation, can stem from primary power source failures, backup power system malfunctions, or issues with the power distribution within the device. Given the Arista access node's role in facilitating data transmission and its reliance on precise hardware and software interoperation, any disruption can significantly impact network performance and availability. Thus, a comprehensive recovery procedure, adhering strictly to Arista's CLI dialect, is essential for diagnosing and rectifying these faults efficiently.

Pre-requisite Checks

- Dispatch remote hands to the Hebbal Edge site for physical inspection and intervention if necessary.

- Verify the status of redundant links to ensure no single point of failure exists.

- Check environmental conditions such as temperature and humidity to ensure they are within the specified operating ranges for the Arista equipment.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Hardware Failure	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

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Command: show hardware tech-support, Expected Output: Detailed hardware and system information., Description: This command provides comprehensive technical support information about the hardware, which can help identify specific component failures or anomalies contributing to the alarm.
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Command: show boot variables, Expected Output: Boot variables and their current settings., Description: Inspecting boot variables can reveal if any recent changes or configurations might have triggered the hardware or environmental fault.

Command: show environment, Expected Output: Environmental sensor readings and thresholds., Description: This command helps in assessing the current environmental conditions (like temperature, fan status) to determine if they are within acceptable ranges for the device's operation.
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Resolution Steps

Step 1: Step: Replace Faulty Line Card, Command: hardware module replace , Description: If the diagnostic process identifies a faulty line card, this step involves physically replacing it with a functional one, ensuring the module number is correctly specified to avoid further complications.

Step 2: Step: Reset ASIC, Command: reset asic , Description: In cases where ASIC memory parity errors are identified, resetting the ASIC can sometimes resolve the issue, though this should be done with caution and only when other diagnostic steps have been exhausted.

Step 3: Step: Restore Power Supply Redundancy, Command: no shutdown , Description: If a power supply redundancy loss is detected, ensuring that all power supplies are operational and configured correctly can mitigate this issue. The specific interface of the power supply to be restored should be identified and the 'no shutdown' command applied accordingly.

Post-Check Verification

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Command: show hardware, Expected Output: Updated hardware status reflecting any changes made., Description: After completing resolution steps, this command verifies that the hardware components are recognized and functioning as expected by the system.

Command: show log, Expected Output: System logs indicating successful recovery or any remaining issues., Description: Inspecting system logs can provide insight into whether the resolution steps were successful and if any further issues need to be addressed.
```

Roll-back Procedures

- Step: Revert Boot Variables, Command: boot-default, Description: If changes made during the resolution process resulted in unforeseen issues, reverting boot variables to their default settings can help restore the system to a known good state.

- Step: Restore Original Line Card, Command: hardware module replace , Description: If a newly installed line card introduces compatibility issues or fails to resolve the original problem, replacing it with the original (if it was not damaged) or another known good module may be necessary.